Diskrete Mathematik – Woche 5

Lucca Chern-Hess
https://github.com/tschai-yim/eth-resources
October 09, 2026

Foundations of Set Theory

Problem 1 ⋆

Prove or disprove for all sets a, B, C. Counterexamples may only use sets built from $\varnothing$.

1. $a \in B \land B \subseteq C \Rightarrow a \in C$
2. $a \in B \land B \in C \Rightarrow a \in C$
3. $a \subseteq B \land B \in C \Rightarrow a \in C$
4. $a \in B \land B \in C \Rightarrow a \subseteq C$
5. $a \in B \Leftrightarrow \{a\} \subseteq B$

Solution:

Only $\subseteq$ "passes elements down". $\in$ goes one nesting level deeper each time.

1. **True**. $B \subseteq C$ means $\forall x(x \in B \to x \in C)$; instantiate with $x = a$.
2. **False**. $a = \varnothing$, $B = \{\varnothing\}$, $C = \{\{\varnothing\}\}$: $a \in B$, $B \in C$, but the only element of C is $\{\varnothing\} \neq \varnothing$, so $a \notin C$.
3. **False**. Same sets: $\varnothing \subseteq \{\varnothing\}$ (Lemma 3.6), $\{\varnothing\} \in C$, but $\varnothing \notin C$.
4. **False**. $a = \{\varnothing\}$, $B = \{\{\varnothing\}\}$, $C = \{\{\{\varnothing\}\}\}$: $a \in B \in C$, but $\varnothing \in a$ and $\varnothing \notin C$, so $a \not\subseteq C$.
5. **True**.

$$\begin{aligned} \{a\} \subseteq B &\Leftrightarrow \forall x(x \in \{a\} \to x \in B) && \text{(Def. 3.3)} \\ &\Leftrightarrow \forall x(x = a \to x \in B) && \text{(only element of } \{a\} \text{ is } a) \\ &\Leftrightarrow a \in B && \text{(the only } x \text{ with } x = a \text{ is } a) \end{aligned}$$

Problem 2 ⋆

Consider the formal proof of Lemma 3.6 ($\varnothing \subseteq A$) in the script.

1. Two steps are only "$\Rightarrow$", not "$\Leftrightarrow$". Why does this not invalidate the proof?
2. Give a direct proof of $\varnothing \subseteq A$ (no contradiction).
3. Prove or disprove: $\forall A(\varnothing \in A)$.
4. Prove or disprove: $\forall A(A \subseteq \varnothing \to A = \varnothing)$.

Solution:

1. A proof by contradiction (Def. 2.17) needs only $\neg S \Rightarrow T$ for some false T. Every $\Leftrightarrow$ step is in particular a $\Rightarrow$ step, so composing all steps (Def. 2.12) gives $\neg S \Rightarrow T$. The converse $T \Rightarrow \neg S$ is never needed.
2. Let A be arbitrary and let x be arbitrary. $x \in \varnothing$ is false (Def. 3.6), so the implication $x \in \varnothing \to x \in A$ is true. As x was arbitrary, $\forall x(x \in \varnothing \to x \in A)$, which is $\varnothing \subseteq A$ by Def. 3.3.
3. **False**. $A = \varnothing$: $\varnothing \notin \varnothing$ by Def. 3.6. Note that $\varnothing \subseteq A$ always holds, but $\varnothing \in A$ does not.
4. **True**. $A \subseteq \varnothing$ and $\varnothing \subseteq A$ (Lemma 3.6), so $A = \varnothing$ by Lemma 3.2.

Problem 3 ⋆

Let $A = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\}$. No justification is required.

1. List all x with $x \in A$ and $x \subseteq A$.
2. List all subsets of A that are **not** elements of A.
3. Compute $|\{\varnothing, \{\varnothing\}, \{\varnothing, \varnothing\}, \{\{\varnothing\}, \varnothing\}, \{\varnothing, \{\varnothing\}\}\}|$.
4. List all elements of $(A \cap \{\varnothing, \{\varnothing, \varnothing\}\}) \times \{\varnothing \cup \{\varnothing\}\}$.
5. Compute $|\mathcal{P}(A \setminus \mathcal{P}(\{\varnothing\}))|$.
6. Compute $|(\varnothing \times A) \cup (\{\varnothing\} \times \{\varnothing\})|$.
7. True or false: $\mathcal{P}(\varnothing) \times \mathcal{P}(\varnothing) = \mathcal{P}(\varnothing \times \varnothing)$.

Solution:

Simplify inner expressions first, then remove duplicate entries ($x \subseteq A$ means every element of x is an element of A).

1. All three elements of A:
   - $\varnothing \subseteq A$ by Lemma 3.6.
   - $\{\varnothing\} \subseteq A$, since its only element $\varnothing$ is in A.
   - $\{\{\varnothing\}\} \subseteq A$, since its only element $\{\varnothing\}$ is in A.
2. A has $2^3 = 8$ subsets. By 1, all three elements of A are among them, so five remain: $\{\{\{\varnothing\}\}\}$, $\{\varnothing, \{\varnothing\}\}$, $\{\varnothing, \{\{\varnothing\}\}\}$, $\{\{\varnothing\}, \{\{\varnothing\}\}\}$, and A itself.
3. 3. The five entries simplify to $\varnothing$, $\{\varnothing\}$, $\{\varnothing\}$, $\{\varnothing, \{\varnothing\}\}$, $\{\varnothing, \{\varnothing\}\}$, which are three distinct sets.
4. $\{\varnothing, \{\varnothing, \varnothing\}\} = \{\varnothing, \{\varnothing\}\}$, and both elements lie in A, so the first factor is $\{\varnothing, \{\varnothing\}\}$. Since $\varnothing \cup \{\varnothing\} = \{\varnothing\}$, the second factor is $\{\{\varnothing\}\}$ (not $\{\varnothing\}$). Result: $\{(\varnothing, \{\varnothing\}), (\{\varnothing\}, \{\varnothing\})\}$.
5. 2. $\mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$ removes two elements of A, so $A \setminus \mathcal{P}(\{\varnothing\}) = \{\{\{\varnothing\}\}\}$ has one element, and a one-element set has $2^1 = 2$ subsets.
6. 1. $\varnothing \times A = \varnothing$ (Example 3.7(i)) and $\{\varnothing\} \times \{\varnothing\} = \{(\varnothing, \varnothing)\}$.
7. **False**. $\mathcal{P}(\varnothing) = \{\varnothing\}$, so the left side is $\{(\varnothing, \varnothing)\}$. The right side is $\mathcal{P}(\varnothing) = \{\varnothing\}$. These differ because the pair $(\varnothing, \varnothing)$ is not the empty set.

Problem 4 ⋆ ⋆

Express as formulas using only $\in$, $=$, logical operators, and quantifiers (no $\varnothing$, $\{\ldots\}$, $\cup$, $\mathcal{P}$, $|\cdot|$):

1. "A is empty"
2. "A has exactly one element"
3. "A has exactly two elements"
4. "$C = A \cup B$"
5. "$P = \mathcal{P}(A)$"

Solution:

1. $\forall x\, \neg(x \in A)$ (Def. 3.6)
2. $\exists x\, \forall y(y \in A \leftrightarrow y = x)$. The "$\leftarrow$" direction says $x \in A$, the "$\rightarrow$" direction says nothing else is in A.
3. $\exists x\, \exists y(\neg(x = y) \land \forall z(z \in A \leftrightarrow (z = x \lor z = y)))$. Without $\neg(x = y)$ this would read "one or two elements".
4. $\forall x(x \in C \leftrightarrow (x \in A \lor x \in B))$ (Def. 3.2 and 3.4)
5. $\forall S(S \in P \leftrightarrow \forall x(x \in S \to x \in A))$ (Def. 3.2, 3.3, 3.7)

Problem 5 ⋆ ⋆

Decide whether each statement is true or false. Give a witness or counterexample, or a short proof.

1. For every set A: $A \cap \mathcal{P}(A) = \varnothing$.
2. There is a non-empty set A with $A \subseteq \mathcal{P}(A)$.
3. For all sets A, B there is a set C with $A \cup C = B$.
4. For all sets A, B there is a set C with $A \setminus C = A \cap B$.
5. There are sets A, B, C with $A \cap B$, $B \cap C$, $A \cap C$ all non-empty but $A \cap B \cap C = \varnothing$.
6. There are pairwise distinct sets A, B, C with $A \cup B = B \cup C = A \cup C$.
7. For all finite sets A, B: $|A \setminus B| = |A| - |B|$.

Solution:

1. **False**. $A = \{\varnothing\}$: $\varnothing \in A$, and $\varnothing \in \mathcal{P}(A)$ because $\varnothing \subseteq A$ (Lemma 3.6). So $\varnothing \in A \cap \mathcal{P}(A)$.
2. **True**. $A = \{\varnothing\}$: its only element $\varnothing$ is a subset of A, i.e. an element of $\mathcal{P}(A)$. Such sets are studied in Problem 10.
3. **False**. $A = \{1\}$, $B = \varnothing$: every C gives $1 \in A \cup C$, so $A \cup C \neq \varnothing = B$.

   Repair: such a C exists iff $A \subseteq B$. If $A \cup C = B$, then $A \subseteq A \cup C = B$. Conversely, if $A \subseteq B$, take $C = B$: $A \cup B = B$ (Theorem 3.4, consistency).
4. **True**. Take $C = A \setminus B$. For any x:

$$\begin{aligned} x \in A \setminus (A \setminus B) &\Leftrightarrow x \in A \land \neg(x \in A \land x \notin B) && \text{(Def. 3.5, twice)} \\ &\Leftrightarrow x \in A \land (x \notin A \lor x \in B) && \text{(de Morgan, double neg.)} \\ &\Leftrightarrow (x \in A \land x \notin A) \lor (x \in A \land x \in B) && \text{(distributivity)} \\ &\Leftrightarrow x \in A \land x \in B && (x \in A \land x \notin A \text{ is false}) \end{aligned}$$

5. **True**. $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{1, 3\}$: $A \cap B = \{2\}$, $B \cap C = \{3\}$, $A \cap C = \{1\}$, but no element lies in all three.
6. **True**. The same three sets are pairwise distinct, and the union of any two of them is $\{1, 2, 3\}$.
7. **False**. $A = \{1\}$, $B = \{2\}$: $|A \setminus B| = 1$, but $|A| - |B| = 0$.

   In general A splits into $A \setminus B$ and $A \cap B$, so $|A \setminus B| = |A| - |A \cap B|$. This equals $|A| - |B|$ iff $|A \cap B| = |B|$, i.e. iff $B \subseteq A$.

Operations on Sets

Problem 6 ⋆

Compute:

1. $\bigcup\{\varnothing\}$
2. $\bigcap\{\varnothing, \{\varnothing\}\}$
3. $\bigcup\{\{\varnothing\}, \{\{\varnothing\}\}\}$
4. $\bigcup\bigcup\{\{\{\varnothing\}\}\}$
5. $\bigcap\{\{\varnothing, \{\varnothing\}\}, \{\{\varnothing\}\}\}$
6. $\bigcup p$ and $\bigcap p$ for $p = \{\{a\}, \{a, b\}\}$

Why does the script not define $\bigcap\varnothing$?

Solution:

Peel off one level of braces: $\bigcup\{A_1, ..., A_k\} = A_1 \cup ... \cup A_k$ and $\bigcap\{A_1, ..., A_k\} = A_1 \cap ... \cap A_k$.

1. $\varnothing$. The only member, $\varnothing$, has no elements.
2. $\varnothing \cap \{\varnothing\} = \varnothing$. Since $\varnothing$ is a member, no x lies in all members.
3. $\{\varnothing\} \cup \{\{\varnothing\}\} = \{\varnothing, \{\varnothing\}\}$.
4. $\{\varnothing\}$. The inner union is $\bigcup\{\{\{\varnothing\}\}\} = \{\{\varnothing\}\}$, then $\bigcup\{\{\varnothing\}\} = \{\varnothing\}$.
5. $\{\varnothing, \{\varnothing\}\} \cap \{\{\varnothing\}\} = \{\{\varnothing\}\}$.
6. $\bigcup p = \{a\} \cup \{a, b\} = \{a, b\}$ and $\bigcap p = \{a\} \cap \{a, b\} = \{a\}$. This also holds if $a = b$.

The script defines $\bigcup\mathcal{A}$ and $\bigcap\mathcal{A}$ only for non-empty $\mathcal{A}$. For $\bigcap$ this restriction is essential: with $\mathcal{A} = \varnothing$, the condition "$x \in A$ for all $A \in \varnothing$" is vacuously true for **every** x. So $\bigcap\varnothing$ would contain every set, and no such set exists. For $\bigcup$ the same formula would just give $\bigcup\varnothing = \varnothing$.

Problem 7 ⋆ ⋆

Prove for all sets A, B, C:

1. $(A \setminus B) \setminus C = A \setminus (B \cup C)$
2. $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$
3. $(A \setminus B) \setminus C = A \setminus (B \setminus C) \Leftrightarrow A \cap C = \varnothing$
4. $A \setminus B = B \setminus A \Leftrightarrow A = B$
5. $A \setminus (A \setminus B) = A \cap B$. For which A, B does $A \setminus (A \setminus B) = B$ hold?

Solution:

1. For any x:

$$\begin{aligned} x \in (A \setminus B) \setminus C &\Leftrightarrow (x \in A \wedge x \notin B) \wedge x \notin C && \text{(Def. 3.5)} \\ &\Leftrightarrow x \in A \wedge \neg(x \in B \vee x \in C) && \text{(de Morgan)} \\ &\Leftrightarrow x \in A \setminus (B \cup C) && \text{(Def. 3.4, 3.5)} \end{aligned}$$

2. For any x:

$$\begin{aligned} x \in A \setminus (B \setminus C) &\Leftrightarrow x \in A \wedge \neg(x \in B \wedge x \notin C) && \text{(Def. 3.5)} \\ &\Leftrightarrow x \in A \wedge (x \notin B \vee x \in C) && \text{(de Morgan, double neg.)} \\ &\Leftrightarrow (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in C) && \text{(distributivity)} \\ &\Leftrightarrow x \in (A \setminus B) \cup (A \cap C) && \text{(Def. 3.4, 3.5)} \end{aligned}$$

3. "⇒": Assume the two sides are equal and let $x \in A \cap C$. By 2, x lies in the right side $A \setminus (B \setminus C)$. But $x \in C$, so $x \notin (A \setminus B) \setminus C$, contradicting the equality. Hence $A \cap C = \varnothing$.

   "⇐": Assume $A \cap C = \varnothing$. Then the right side is $(A \setminus B) \cup \varnothing = A \setminus B$ by 2. On the left, no $x \in A \setminus B$ lies in C (it lies in A), so removing C changes nothing: $(A \setminus B) \setminus C = A \setminus B$. Both sides equal $A \setminus B$.
4. "⇐": if $A = B$, both sides are $A \setminus A = \varnothing$.

   "⇒": Assume $A \setminus B = B \setminus A$. Suppose some $x \in A \setminus B$. Then $x \in B \setminus A$, so $x \in B$, contradicting $x \notin B$. Hence $A \setminus B = \varnothing$, i.e. every $x \in A$ lies in B: $A \subseteq B$. Symmetrically $B \subseteq A$, so $A = B$ (Lemma 3.2).
5. Proved in Problem 5.4. Then $A \setminus (A \setminus B) = B$ iff $A \cap B = B$ iff $B \subseteq A$ (Theorem 3.4, consistency and commutativity). So "removing twice" only gives back B when B lies inside A.

Problem 8 ⋆ ⋆

Using only $\cup$, $\cap$, $\setminus$ and parentheses, write each set as an expression in which A, B, C each appear at most once.

1. $(A \setminus B) \cup (A \setminus C)$
2. $(A \cup C) \setminus (B \cup C)$
3. $(A \cap B) \cup (A \setminus C)$
4. Prove that $(A \setminus B) \cup (B \setminus A)$ **cannot** be written with A and B each appearing at most once.

Solution:

Write a, b, c for $x \in A$, $x \in B$, $x \in C$. Translate the set into a formula, simplify it with Lemma 2.1, then read it back as a set ($\wedge$ is $\cap$, $\vee$ is $\cup$, $F \wedge \neg G$ is $\setminus$).

1. Distributivity backwards, then de Morgan:
$$(a \wedge \neg b) \vee (a \wedge \neg c) \equiv a \wedge (\neg b \vee \neg c) \equiv a \wedge \neg(b \wedge c),$$
so the set is $A \setminus (B \cap C)$.
2. Distributivity, then drop the false term $c \wedge \neg c \wedge \neg b$:
$$(a \vee c) \wedge \neg b \wedge \neg c \equiv (a \wedge \neg b \wedge \neg c) \vee (c \wedge \neg c \wedge \neg b) \equiv a \wedge \neg b \wedge \neg c \equiv a \wedge \neg(b \vee c),$$
so the set is $A \setminus (B \cup C)$. Note that C disappears from the "positive" part.
3. Distributivity backwards, then de Morgan and double negation:
$$(a \wedge b) \vee (a \wedge \neg c) \equiv a \wedge (b \vee \neg c) \equiv a \wedge \neg(c \wedge \neg b),$$
so the set is $A \setminus (C \setminus B)$. This is Problem 7.2 with B and C swapped.
4. An expression with only two set symbols is a single symbol or one operation applied to two symbols. Since $\cup$ and $\cap$ are commutative, the only candidates are A, B, $A \cup B$, $A \cap B$, $A \setminus B$, $B \setminus A$. Each one differs from the symmetric difference $D = (A \setminus B) \cup (B \setminus A)$ for some choice of sets:
   - $A = B = \{1\}$: $D = \varnothing$, but $A = B = A \cup B = A \cap B = \{1\}$.
   - $A = \{1\}$, $B = \varnothing$: $D = \{1\}$, but $B \setminus A = \varnothing$.
   - $A = \varnothing$, $B = \{1\}$: $D = \{1\}$, but $A \setminus B = \varnothing$.

Problem 9 ⋆ ⋆

Let A, B be sets and $\mathcal{A}$ a set of sets.

1. Prove: $\bigcup \mathcal{P}(A) = A$.
2. Prove: $\mathcal{A} \subseteq \mathcal{P}(\bigcup \mathcal{A})$. Give an example where the inclusion is strict.
3. Prove: $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.
4. Prove or disprove: $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.

Solution:

1. "$\subseteq$": Let $x \in \bigcup \mathcal{P}(A)$. Then $x \in S$ for some $S \in \mathcal{P}(A)$, i.e. some $S \subseteq A$. Hence $x \in A$.

   "$\supseteq$": Let $x \in A$. Since $A \subseteq A$, we have $A \in \mathcal{P}(A)$, so x lies in a member of $\mathcal{P}(A)$, i.e. $x \in \bigcup \mathcal{P}(A)$.

   Lemma 3.2 gives equality.
2. Let $X \in \mathcal{A}$. Every $x \in X$ lies in a member of $\mathcal{A}$, so $x \in \bigcup \mathcal{A}$. Hence $X \subseteq \bigcup \mathcal{A}$, i.e. $X \in \mathcal{P}(\bigcup \mathcal{A})$.

   Strict: $\mathcal{A} = \{\{1\}, \{2\}\}$ gives $\mathcal{P}(\bigcup \mathcal{A}) = \mathcal{P}(\{1,2\}) = \{\varnothing, \{1\}, \{2\}, \{1,2\}\} \neq \mathcal{A}$.
3. For any S:

$$\begin{aligned} S \in \mathcal{P}(A \cap B) &\Leftrightarrow \forall x(x \in S \to (x \in A \wedge x \in B)) && \text{(Def. 3.7, 3.3, 3.4)} \\ &\Leftrightarrow \forall x((x \in S \to x \in A) \wedge (x \in S \to x \in B)) && (*) \\ &\Leftrightarrow \forall x(x \in S \to x \in A) \wedge \forall x(x \in S \to x \in B) && (**) \\ &\Leftrightarrow S \in \mathcal{P}(A) \cap \mathcal{P}(B) && \text{(Def. 3.3, 3.7, 3.4)} \end{aligned}$$

$(*)$: $F \to (G \wedge H) \equiv (F \to G) \wedge (F \to H)$, checked with a truth table. $(**)$: $\forall x\,(F \wedge G) \equiv \forall x\,F \wedge \forall x\,G$ (script, Section 2.4.8).

4. **False**. $A = \{1\}$, $B = \{2\}$: $\{1,2\} \subseteq A \cup B$, so $\{1,2\} \in \mathcal{P}(A \cup B)$. But $\{1,2\}$ is a subset of neither A nor B.

   Only "$\supseteq$" holds in general: $S \subseteq A$ or $S \subseteq B$ implies $S \subseteq A \cup B$.

   Equality holds iff $A \subseteq B$ or $B \subseteq A$. If $A \subseteq B$, then $A \cup B = B$ and $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, so both sides are $\mathcal{P}(B)$ (symmetrically for $B \subseteq A$). If neither holds, pick $a \in A \setminus B$ and $b \in B \setminus A$. Then $\{a,b\} \subseteq A \cup B$, but $\{a,b\} \not\subseteq A$ (as $b \notin A$) and $\{a,b\} \not\subseteq B$ (as $a \notin B$).

Problem 10 ⋆ ⋆

Call a set T *transitive* if $x \in T \Rightarrow x \subseteq T$.

1. Which of $\{\varnothing\}$, $\{\{\varnothing\}\}$, $\{\varnothing, \{\{\varnothing\}\}\}$, $\{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\}$ are transitive?
2. Prove: T is transitive $\Leftrightarrow T \subseteq \mathcal{P}(T) \Leftrightarrow \bigcup T \subseteq T$.
3. Prove: if T is transitive, then $\mathcal{P}(T)$ is transitive.
4. Prove: if T is transitive, then $T \cup \{T\}$ is transitive.

Solution:

To check transitivity, go through each element $x \in T$ and check that every element of x is again in T.

1. • $\{\varnothing\}$: transitive. Its only element $\varnothing$ is a subset of everything (Lemma 3.6).
   • $\{\{\varnothing\}\}$: not transitive. $\{\varnothing\}$ is an element, but its element $\varnothing$ is not in $\{\{\varnothing\}\}$.
   • $\{\varnothing, \{\{\varnothing\}\}\}$: not transitive. $\{\{\varnothing\}\}$ is an element, but its element $\{\varnothing\}$ is not in the set.
   • $\{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}\}$: transitive, see Problem 3.1.
2. "Transitive $\Leftrightarrow T \subseteq \mathcal{P}(T)$": by Def. 3.7, $x \subseteq T$ means $x \in \mathcal{P}(T)$. So the definition "every $x \in T$ satisfies $x \subseteq T$" reads "every $x \in T$ satisfies $x \in \mathcal{P}(T)$", which is $T \subseteq \mathcal{P}(T)$ (Def. 3.3).

   "Transitive $\Rightarrow \bigcup T \subseteq T$": Let $y \in \bigcup T$. Then $y \in x$ for some $x \in T$. As T is transitive, $x \subseteq T$, so $y \in T$.

   "$\bigcup T \subseteq T \Rightarrow$ transitive": Let $x \in T$; we show $x \subseteq T$. Let $y \in x$. Then y lies in the member x of T, so $y \in \bigcup T \subseteq T$.

   (For $T = \varnothing$ all three statements hold trivially, reading $\bigcup \varnothing$ as $\varnothing$.)
3. Let $x \in \mathcal{P}(T)$, i.e. $x \subseteq T$. We show $x \subseteq \mathcal{P}(T)$. Let $y \in x$. Then $y \in T$ (as $x \subseteq T$), so $y \subseteq T$ since T is transitive, i.e. $y \in \mathcal{P}(T)$.
4. Let $x \in T \cup \{T\}$. We show $x \subseteq T \cup \{T\}$.
   • Case $x \in T$: $x \subseteq T$ since T is transitive, and $T \subseteq T \cup \{T\}$. By Lemma 3.3, $x \subseteq T \cup \{T\}$.
   • Case $x \in \{T\}$, i.e. $x = T$: $T \subseteq T \cup \{T\}$.

Problem 11 ⋆ ⋆ ⋆

Let A, B, C be sets.

1. Prove: there is a set X with $A \cup X = B$ and $A \cap X = C$ if and only if $C \subseteq A \subseteq B$. Show that in this case X is unique, and give it explicitly.

For sets S, T let $S \triangle T = (S \setminus T) \cup (T \setminus S)$ (the *symmetric difference*).

2. Prove that $\triangle$ is associative: $(A \triangle B) \triangle C = A \triangle (B \triangle C)$.
3. Prove: for all A, B there is **exactly one** set X with $A \triangle X = B$.

Solution:

1. "⇒": Let X be a solution. Then $C = A \cap X \subseteq A$ and $A \subseteq A \cup X = B$.

   Uniqueness. Let X be a solution and x arbitrary.
   - $x \in A$: then $x \in X \Leftrightarrow x \in A \cap X \Leftrightarrow x \in C$.
   - $x \notin A$: then $x \in X \Leftrightarrow x \in A \cup X \Leftrightarrow x \in B$.

   So $x \in X \Leftrightarrow (x \in A \wedge x \in C) \vee (x \notin A \wedge x \in B)$, i.e. $X = (A \cap C) \cup (B \setminus A)$. If $C \subseteq A$, this is $X = C \cup (B \setminus A)$ (Theorem 3.4, consistency). Every solution equals this set, so there is at most one solution.

   "⇐": Assume $C \subseteq A \subseteq B$ and let $X = C \cup (B \setminus A)$. We check both equations.
   - $A \cup X = A \cup C \cup (B \setminus A) = A \cup (B \setminus A)$, since $C \subseteq A$. For any x:

     $$x \in A \vee (x \in B \wedge x \notin A) \equiv (x \in A \vee x \in B) \wedge (x \in A \vee x \notin A) \equiv x \in A \vee x \in B$$

     by distributivity, and because $x \in A \vee x \notin A$ is true. So $A \cup X = A \cup B = B$, since $A \subseteq B$.
   - $A \cap X = (A \cap C) \cup (A \cap (B \setminus A)) = C \cup \varnothing = C$, by distributivity, $C \subseteq A$, and since no element of $B \setminus A$ lies in A.

   The trap: $X = B \setminus A$ satisfies the first equation, but $A \cap (B \setminus A) = \varnothing$, so it only works for $C = \varnothing$.
2. Write a, b, c for $x \in A$, $x \in B$, $x \in C$, and $a \oplus b$ for $(a \wedge \neg b) \vee (\neg a \wedge b)$ ("exactly one of a, b"). By Def. 3.4 and 3.5, $x \in A \triangle B \Leftrightarrow a \oplus b$. Hence

   $$x \in (A \triangle B) \triangle C \Leftrightarrow (a \oplus b) \oplus c \quad \text{and} \quad x \in A \triangle (B \triangle C) \Leftrightarrow a \oplus (b \oplus c).$$

   $p \oplus q$ is true iff exactly one of p, q is true. So $a \oplus b$ is true iff an odd number of a, b is true, and $(a \oplus b) \oplus c$ is true iff an odd number of a, b, c is true. The same holds for $a \oplus (b \oplus c)$. The formulas are equivalent (one can also check all 8 rows of the truth table), so both sets have the same elements (Def. 3.2).
3. Since $a \oplus a \equiv 0$ and $0 \oplus b \equiv b$, we have $A \triangle A = \varnothing$ and $\varnothing \triangle B = B$.

   Existence. By 2, $A \triangle (A \triangle B) = (A \triangle A) \triangle B = \varnothing \triangle B = B$, so $X = A \triangle B$ works.

   Uniqueness. If $A \triangle X = B$, then by 2

   $$X = \varnothing \triangle X = (A \triangle A) \triangle X = A \triangle (A \triangle X) = A \triangle B.$$

   Unlike for $\cup$ (Problem 5.3), a solution always exists: applying $A \triangle \cdot$ again "undoes" A, like subtraction undoes $+$.

Ordered Pairs and Cartesian Products

Problem 12 ⋆ ⋆

Prove for all sets A, B:

1. $A \times B = B \times A \Leftrightarrow A = B \vee A = \varnothing \vee B = \varnothing$
2. $A \times A = B \times B \Rightarrow A = B$

Solution:

1. "$\Leftarrow$": If $A = B$, both sides are $A \times A$. If $A = \varnothing$ or $B = \varnothing$, both sides are $\varnothing$, since no pair has a component in $\varnothing$ (Example 3.7(i)).

   "$\Rightarrow$": Assume $A \times B = B \times A$. If $A = \varnothing$ or $B = \varnothing$, we are done. Otherwise fix $a_0 \in A$ and $b_0 \in B$.
   - Let $x \in A$. Then $(x, b_0) \in A \times B = B \times A$, so its first component satisfies $x \in B$. Hence $A \subseteq B$.
   - Let $y \in B$. Then $(a_0, y) \in A \times B = B \times A$, so its second component satisfies $y \in A$. Hence $B \subseteq A$.
   - Lemma 3.2 gives $A = B$.
2. Let $x \in A$. Then $(x, x) \in A \times A = B \times B$, so $x \in B$. Hence $A \subseteq B$, symmetrically $B \subseteq A$, and $A = B$ by Lemma 3.2.

   Unlike in 1, the pair (x, x) is built from x alone, so no element of the other set is needed and no case $A = \varnothing$ appears. (For $A = \varnothing$ the argument holds vacuously.)

Problem 13 ⋆ ⋆

Prove or disprove. Repair false claims with an additional hypothesis.

1. $A \times B \subseteq C \times D \Rightarrow A \subseteq C$
2. $A \times B = A \times C \Rightarrow B = C$
3. $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$
4. $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$
5. Let A, B be finite with $|A| = m$, $|B| = n$, $|A \cap B| = k$. Compute $|(A \times B) \cup (B \times A)|$.

Solution:

1. **False**. $A = \{1\}$, $B = C = D = \varnothing$: $A \times B = \varnothing \subseteq C \times D$ (Lemma 3.6), but $A \not\subseteq C$.

   Repair: add $B \neq \varnothing$. Fix $b_0 \in B$. For $x \in A$, $(x, b_0) \in A \times B \subseteq C \times D$, so $x \in C$.
2. **False**. $A = \varnothing$, $B = \{1\}$, $C = \{2\}$: both products are $\varnothing$, but $B \neq C$.

   Repair: add $A \neq \varnothing$. Fix $a_0 \in A$. For $y \in B$, $(a_0, y) \in A \times B = A \times C$, so $y \in C$. Hence $B \subseteq C$, symmetrically $C \subseteq B$.
3. **False**. $A = B = \{1\}$, $C = D = \{2\}$: the left side is $\{(1,1),(2,2)\}$, but the right side $\{1,2\} \times \{1,2\}$ also contains $(1,2)$.

   "$\subseteq$" always holds: $(x, y) \in A \times B$ gives $x \in A \cup C$ and $y \in B \cup D$, and likewise for $C \times D$.

   Repair: e.g. $A = C$. Then both sides are $A \times (B \cup D)$, because $x \in A \wedge (y \in B \vee y \in D) \equiv (x \in A \wedge y \in B) \vee (x \in A \wedge y \in D)$ (distributivity). The case $B = D$ works the same way.
4. **True**.

$$\begin{aligned}(x,y) \in (A \times B) \cap (C \times D) &\Leftrightarrow x \in A \wedge y \in B \wedge x \in C \wedge y \in D && \text{(Def. 3.4, 3.8)}\\ &\Leftrightarrow (x \in A \cap C) \wedge (y \in B \cap D) && \text{(comm., assoc.)}\\ &\Leftrightarrow (x,y) \in (A \cap C) \times (B \cap D) && \text{(Def. 3.8)}\end{aligned}$$

5. $2mn - k^2$. By 4 (with $C = B$, $D = A$), $(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$, which has $k \cdot k = k^2$ elements. Adding the sizes of both products counts the pairs in the intersection twice, so

$$|(A \times B) \cup (B \times A)| = mn + nm - k^2 = 2mn - k^2.$$

Check: $A = B$ gives $m = n = k$ and $m^2 = |A \times A|$; disjoint A, B give $2mn$.

Problem 14 ⋆ ⋆

1. List $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$ for $A = \{1\}$, $B = \{2\}$.
2. For finite A, B with $|A| = m$, $|B| = n$: for which m, n do $\mathcal{P}(A \times B)$ and $\mathcal{P}(A) \times \mathcal{P}(B)$ have the same cardinality?
3. Prove: $\mathcal{P}(A \times B) \neq \mathcal{P}(A) \times \mathcal{P}(B)$ for all sets A, B.

Solution:

1. $A \times B = \{(1,2)\}$, so $\mathcal{P}(A \times B) = \{\varnothing, \{(1,2)\}\}$ is a set of **sets of pairs**. In contrast, $\mathcal{P}(A) \times \mathcal{P}(B) = \{(\varnothing, \varnothing), (\varnothing, \{2\}), (\{1\}, \varnothing), (\{1\}, \{2\})\}$ is a set of **pairs of sets**.
2. $m = n = 0$ or $m = n = 2$. We have $|\mathcal{P}(A \times B)| = 2^{mn}$ and $|\mathcal{P}(A) \times \mathcal{P}(B)| = 2^m \cdot 2^n = 2^{m+n}$, so

$$\begin{aligned} 2^{mn} = 2^{m+n} &\Leftrightarrow mn = m + n \\ &\Leftrightarrow mn - m - n + 1 = 1 \\ &\Leftrightarrow (m-1)(n-1) = 1 \\ &\Leftrightarrow m - 1 = n - 1 = 1 \vee m - 1 = n - 1 = -1 \quad (\text{integer factors of } 1 \text{ are } \pm 1) \end{aligned}$$

3. $\varnothing \subseteq A \times B$ (Lemma 3.6), so $\varnothing \in \mathcal{P}(A \times B)$. Every element of $\mathcal{P}(A) \times \mathcal{P}(B)$ is an ordered pair, and $\varnothing$ is not an ordered pair. So $\varnothing \notin \mathcal{P}(A) \times \mathcal{P}(B)$, and the two sets differ. This holds even for $A = B = \varnothing$: then $\mathcal{P}(A \times B) = \{\varnothing\}$ and $\mathcal{P}(A) \times \mathcal{P}(B) = \{(\varnothing, \varnothing)\}$.